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Course Name – CSE AI or ML

Project Title: Movie Recommendation System

Subject: Artificial Intelligence and Machine learning

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🎬 Movie Recommendation System — Project Report

1. Introduction

In today's digital world, users are overwhelmed with a vast number of movie choices available on various platforms. Choosing what to watch often becomes a time-consuming task. Recommendation systems help solve this problem by suggesting relevant content based on user preferences.

This project focuses on building a **Movie Recommendation System** using a content-based filtering approach. It recommends movies similar to a given movie by analyzing their features such as overview and genres.

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2. Problem Statement

The main problem addressed in this project is:

How can we suggest movies to users based on their interests without relying on user ratings or history?

Many platforms use complex systems, but this project aims to create a **simple and effective recommendation system** using basic machine learning techniques.

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3. Why This Problem Matters

- Users often struggle to find movies they will enjoy
- Too many choices lead to decision fatigue
- Personalized recommendations improve user experience
- Recommendation systems are widely used in platforms like Netflix and Amazon

This project demonstrates how such systems work at a basic level.

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4. Approach

This project uses a **content-based filtering technique**, which recommends movies based on similarity between their content.

Steps followed:

1. Data collection from a movie dataset
2. Data preprocessing (handling missing values, selecting important columns)
3. Feature engineering (combining overview and genres)
4. Text vectorization using TF-IDF

5. Similarity calculation using cosine similarity
6. Building a recommendation function
7. Creating a user interface using Streamlit

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5. Implementation

5.1 Dataset

The dataset used contains:

- Movie Title
- Overview (plot summary)

5.2 Data Preprocessing

- Removed null values
- Selected relevant columns
- Cleaned text data

5.3 Feature Engineering

- Combined overview and genres into a single column
- This helps the model understand both story and type of movie

5.4 TF-IDF Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) converts text into numerical vectors so that machine learning algorithms can process it.

5.5 Cosine Similarity

Cosine similarity measures how similar two movies are based on their vector representations.

5.6 Recommendation Function

- Takes a movie name as input
- Finds similar movies using similarity scores
- Returns top recommended movies

5.7 User Interface

- Built using Streamlit
- Allows users to input a movie name
- Displays recommended movies instantly

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6. Key Decisions

- Used **content-based filtering** instead of collaborative filtering (simpler and no user data required)
- Chose **TF-IDF** for text processing (efficient and widely used)
- Used **Streamlit** for quick and easy UI development
- Combined multiple features for better recommendation accuracy

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7. Challenges Faced

- Understanding how recommendation systems work initially
- Handling text data properly
- Cleaning and formatting dataset columns
- Fixing incorrect recommendations due to poor feature selection
- Debugging issues like missing columns and indexing errors

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8. Results

After improvements, the system is able to:

- Recommend movies with similar genres and themes
- Provide relevant suggestions for popular movies
- Deliver results quickly through a simple interface

Example:
Input: *Avatar*
Output: Similar science fiction and action movies

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9. Learning Outcomes

Through this project, I learned:

- Basics of machine learning and recommendation systems
- How TF-IDF and cosine similarity work
- Importance of feature selection
- Data preprocessing techniques

- Building real-world projects using Python
- Creating interactive apps using Streamlit

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10. Conclusion

This project successfully demonstrates a basic movie recommendation system using content-based filtering. While simple, it reflects the core idea behind real-world recommendation engines.

The system can be further improved by adding more features such as cast, director, and user preferences.

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11. Future Improvements

- Add movie posters and better UI
- Include cast and director information
- Use larger and more detailed datasets
- Implement hybrid recommendation systems
- Deploy the application online